Exploring human representations of facial affect by integrating a deep generative model into Markov Chain Monte Carlo With People

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Introduction

People's internal representations of natural categories can be viewed as density over a set of objects, as described in classic *examplar*[1] or *prototype model*[2], which is crucial for people to perceive, learn, and infer:

One of the most powerful methods for estimating such representations is Markov Chain Monte Carlo with People (MCMCP), in which people serve as the acceptance function of the Metropolis-Hastings algorithm[3] and perform consecutive pairwise decisions to sample from their category representations[4] (*Fig.1*). However, **MCMCP requires a large number of trials to converge, particularly for high-dimensional stimuli such as faces**.

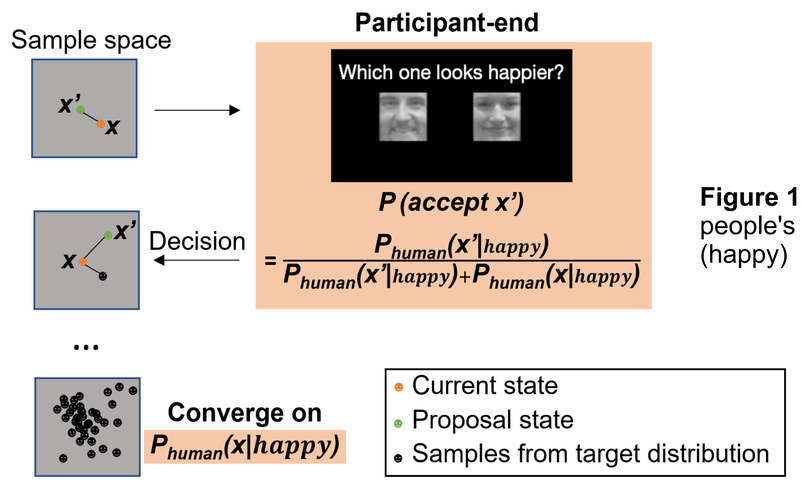

Figure 1. Using MCMCP to identify people's representation of facial affect (happy)

Variational Auto-Encoder (VAE)[5] is considered a successful generative model, which does well in uncovering the latent structure underlying observations. Accordingly, in the study we proposed and tested a VAE-guided approach to **navigate MCMCP by using the implicit knowledge of natural categories that VAE learned from image datasets**.

Method

- We ran a new experiment applying our VAE-guided MCMCP to recovering people's representations of happy and sad faces;

- The VAE was modified by adding a classifier following the latent layer (*Fig.2*) to enable the conditional generation of facial affect. The model was trained using the California Facial Expression (CAFE) dataset;

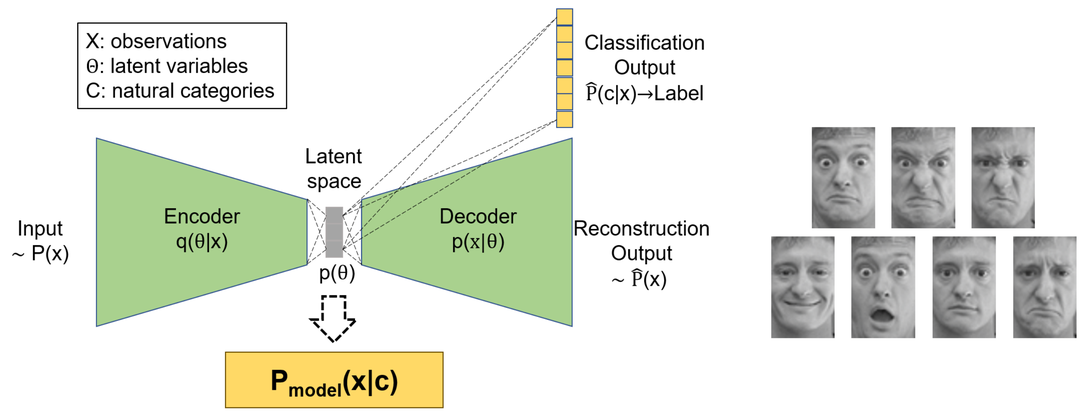

Figure 2. Modified Variational Auto-Encoder (left) and example faces from the CAFE (right)

- The initial states of Markov chains were selected uniformly within the high-density area of $P_{model}(x|c)$, where c refers to happy or sad;

- Only proposals that were jointly accepted by participants and VAE can be counted as new samples from the target distribution (*Fig.3*).

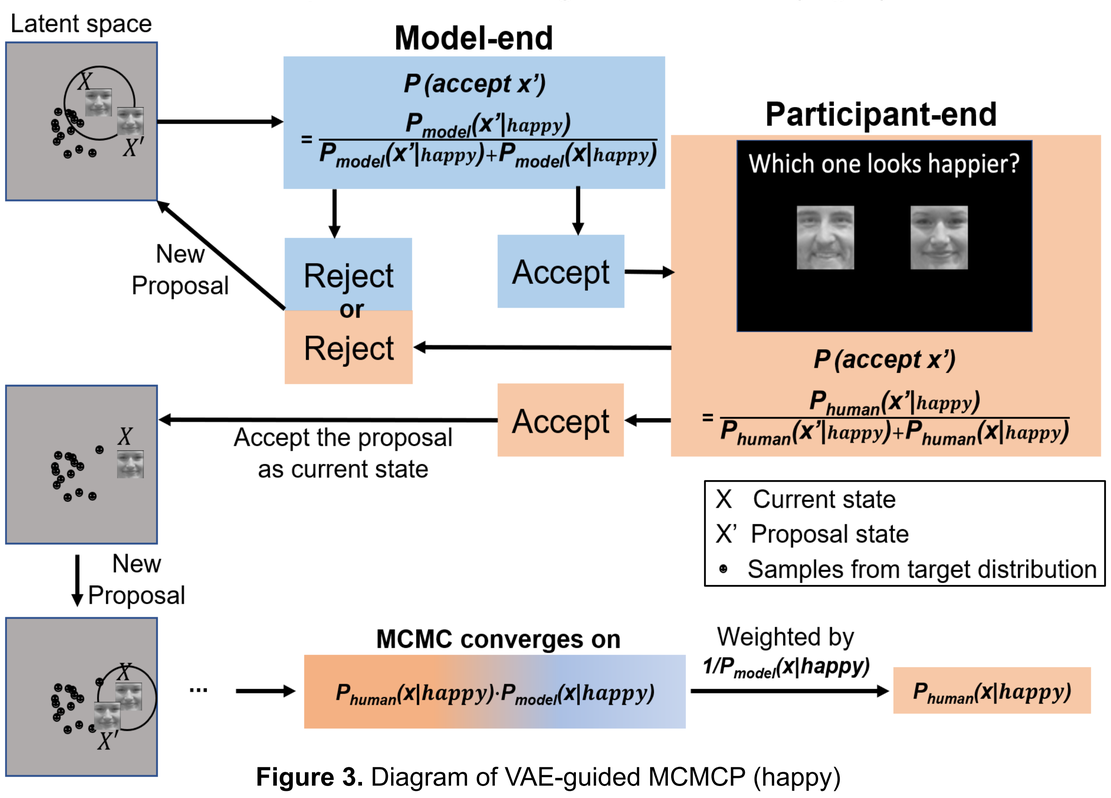

Figure 3. Diagram of VAE-guided MCMCP (happy)

Results

- The 3-d latent space witnessed seven clusters of facial affect after training, corresponding to seven categories of facial affect (*Fig.4a*).

- Samples acquired by a participant were presented in *Fig.4b* (bubble size indiciates sample size at the coordinates). Average acceptance rate of VAE (i.e., percentage of trials performed by people) were around 20%.

- Variance ratio indicated the convergence of Markov Chains for most participants (*Fig.4c*).

- The weighted samples from $P_{human}(x|c)$ were averaged within individuals, and visualised in *Fig.4d*. Significant individual differences were observed, particularly in sad faces (participants 1, 9 and 17 were shown in *Fig.4e*).

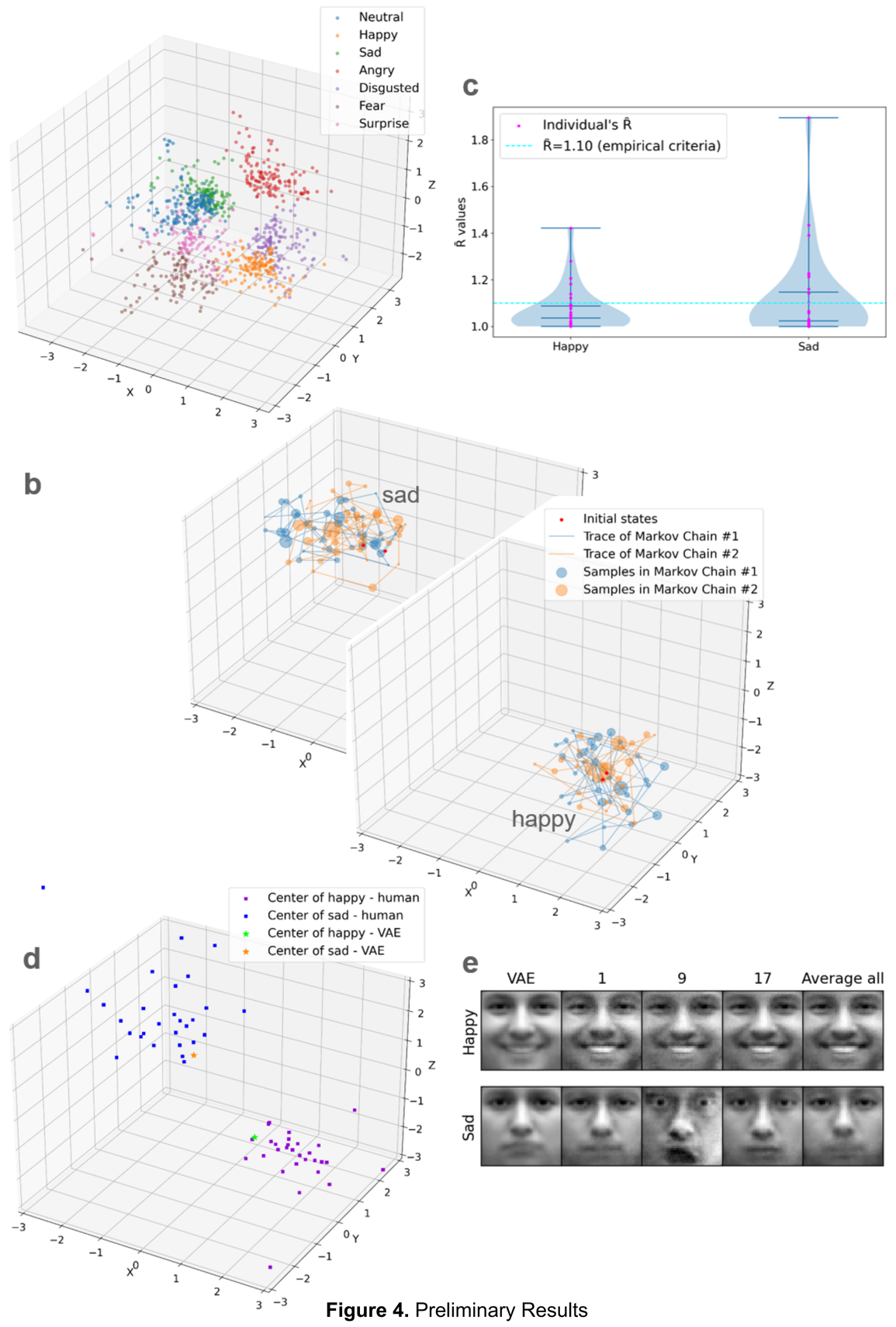

Figure 4. Preliminary Results

Conclusion and further study

- Our method could effectively accelerate MCMCP, making it possible to identify people's categorical representations at the individual level;

- Participants show greater individual differences in their representations for sad facial affect than that for happy;

- More fine-grained evaluation and application of VAE-guided MCMCP need to be conducted following the study.

Reference:

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Source code: **Lab:**